gasification
followed by
methanation and
liquefaction
Grid mix electricity LNG_b_G_M_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
33 LNG Liquefied
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion,

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SOURCE: MEPC.376(80).pdf
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
33 LNG Liquefied
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion,
separation and
liquefaction
Grid mix electricity LNG_b_AD_gm
34 LNG Liquefied
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.

feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion,
separation with
Point Source
Carbon Capture
(PSCC) and long-
term storage and
liquefaction
Grid mix electricity LNG_b_AD_CCS_gm
35 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Methanation and
liquefaction
Grid mix electricity LNG_fCO2_fH2_M_gm
36 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: from
Renewable
electricity
Methanation and
liquefaction
Grid mix electricity LNG_fCO2_rH2_M_gm

Order Group Fuel type

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SOURCE: MEPC.376(80).pdf
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 32

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Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
37 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_fCO2_ibpH2_M_gm
38 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Fossil Steam
Methane
Reformation
Methanation and
liquefaction
Grid mix electricity LNG_rCO2_fH2_M_gm
39 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: from
Renewable
electricity
Methanation and
liquefaction
Grid mix electricity LNG_rCO2_rH2_M_gm
40 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_rCO2_ibpH2_M_gm
41 LNG Liquefied

Natural Gas
(Methane)

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SOURCE: MEPC.376(80).pdf
40 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_rCO2_ibpH2_M_gm
41 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture